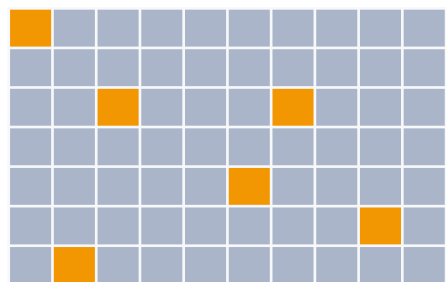


Example weight tensor



- Train only these weights
- Frozen weights



## Sparse High Rank Adapters (SHiRA)

Finetune a small fraction of the base model's parameters for the adaptation tasks and store them as only the sparse weights and their indices



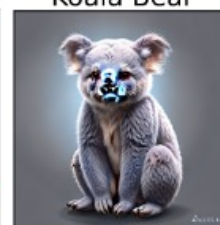
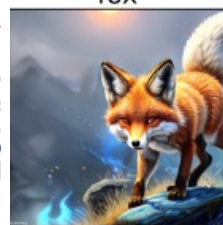
Bluefire Adapter



Paintings Adapter

Multi-Adapter  
Fusion

LoRA  
SHiRA



Bluefire is lost



Artifacts for unknown classes during training



Paintings is lost



SHiRA significantly reduces concept-loss compared to LoRA when multiple adapters are combined. In general, SHiRA performs better than LoRA on a variety of vision and language tasks.